Humanoid VLA models typically use pretrained vision–language models (VLMs) or video generation models (VGMs). VLM-based methods transfer visual-semantic knowledge to actions (Bjorck et al., 2025; Wei et al., 2026a; Zhong et al., 2025; Hu et al., 2026): GR00T N1 (Bjorck et al., 2025) couples a VLM with a diffusion action model, while Ψ_0 (Wei et al., 2026a) transfers representations from human egocentric demonstrations to a humanoid action expert. VGM-based methods exploit visual dynamics priors (Zheng et al., 2026; Li et al., 2026): MotionWAM (Zheng et al., 2026) generates whole-body motion from intermediate video features; WholeBodyWAM (Li et al., 2026) grounds world–action priors in whole-body controllers. Unified generators (Hu et al., 2026; Zheng et al., 2026) share prediction across motion components; hybrids (Zhong et al., 2025; Jiang et al., 2025) combine upper-body targets with locomotion commands. HAF (Gu et al., 2026) stages generation with a shared action expert, expanding the active action set and refining earlier predictions. Kimodo-Policy instead transfers a T2M backbone with pretrained inter-joint and temporal priors, reducing the need to learn motion regularities from humanoid demonstrations. Dedicated root, body, and hand generators meet distinct control requirements, while cross-stage conditioning supports coordinated, visually grounded whole-body behavior.

T2M generation evolved from TEMOS's variational language–motion modeling (Petrovich et al., 2022) to diffusion and quantized tokens. MDM (Tevet et al., 2023) and MLD (Chen et al., 2023) model pose and latent distributions, respectively; T2M-GPT (Zhang et al., 2023a) and MoMask (Guo et al., 2024) use autoregressive and masked token modeling, respectively. OmniControl (Xie et al., 2024) supports spatial constraints; Kimodo (Rempe et al., 2026) scales controllable hierarchical root–body diffusion through large-scale training for human and humanoid skeletons. Humanoid-VLA (Ding et al., 2025) and ZeroWBC (Yang et al., 2026) learn autoregressive motion priors within pretrained language and vision–language models, respectively, before visual integration or task-specific adaptation. MotionWAM (Zheng et al., 2026) adapts a pretrained video world model through staged video–motion training; WholeBodyWAM (Zhang et al., 2026) pretrains a motion expert and integrates it with video and action experts. Kimodo-Policy directly repurposes a diffusion-based T2M generator pretrained on large-scale motion data in robot kinematic space as its backbone. Its root–body–hand generators use vision and motion history for closed-loop loco-manipulation, transferring inter-joint and temporal priors to scene-dependent actions, reducing relearning from robot demonstrations, and accommodating component-specific control requirements.

- We introduce **Kimodo-Policy**, a humanoid VLA framework transferring pretrained T2M priors to whole-body control. Pretraining on diverse humanoid demonstrations conditions closed-loop actions on visual observations, language instructions, and motion history.
- We propose dedicated root, body, and hand generators in a three-stage architecture to meet distinct control requirements. Explicit cross-stage conditioning preserves motion dependencies, supporting specialized motion generation and coordinated whole-body control.
- We demonstrate strong performance on HumanoidArena, SIMPLE, and real-world humanoid tasks using 678M parameters (only 92M trainable), with consistent gains from larger VLA pretraining datasets, greater model capacity, and stronger T2M priors.